

Preparing for BIO 340 General Genetics

A refresher using material from introductory biology (BIO 181 and BIO 182)

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Welcome to BIO 340. If you would like to prepare before class starts, begin with material from your introductory biology classes (BIO 181 and BIO 182 at ASU). We will build on the material you learned in those courses throughout General Genetics.

We will review several of these topics in BIO 340 as we use them, but a little review now can make the start of the course easier. Genetics builds from one idea to the next, so getting behind can make later material harder. Reviewing now gives you a stronger starting point and will make it easier to keep up as the course progresses. You are in a good place if you can use the material and explain it without notes or reference material.

What is worth reviewing

- Work comfortably with fractions, decimals, percentages, ratios, and simple probability.
- Explain the relationships among DNA, RNA, proteins, genes, chromosomes, and traits.
- Distinguish mitosis from meiosis both by description and using images.
- Describe replication, transcription, and translation at the introductory biology level.
- Connect chemical bonds and molecular interactions to DNA structure and protein folding.

Mathematical tools you will use

You will use basic arithmetic to solve genetics problems. It helps to be comfortable with these skills and to know why you are using each one.

- Add, subtract, multiply, and simplify fractions. Convert among fractions, decimals, and percentages.
- Review the product rule for independent events.
- Review the sum rule for mutually exclusive outcomes.
- Recognize ratios and proportions, use exponents, and rearrange a simple equation.
- Read a graph or table, identify the variables, compare groups, and describe the pattern without overinterpreting it.

A few examples

- $1/2 \times 1/2 = 1/4$
- $1/4 + 1/4 = 1/2$
- $0.07 = 7\%$
- $3/8 = 0.375 = 37.5\%$

Chemistry and macromolecules

Review the bond types and molecular interactions that are important for DNA, RNA, and protein structure.

Interaction	Related ideas to review
Covalent bonds	Peptide, phosphodiester, glycosidic, and disulfide bonds
Hydrogen bonds	Complementary base pairing and protein secondary structure
Ionic interactions	Charged side chains, salt bridges, and the effects of pH
van der Waals interactions	Molecular shape and stability
Hydrophobic effect	Protein folding and membrane formation

Also review electronegativity, polar versus nonpolar bonds, partial and full charges, water as a solvent, pH, and common functional groups such as hydroxyl, carbonyl, carboxyl, amino, phosphate, sulfhydryl, and methyl groups.

Monomers polymers and protein properties

- Identify nucleotide monomers and nucleic acid polymers. Identify amino acid monomers and polypeptide or protein polymers.
- Explain dehydration synthesis and hydrolysis at a conceptual level.
- Recognize an amino acid's amino group, carboxyl group, central carbon, hydrogen, and variable R group.
- Use an R group's structure to classify it as nonpolar, polar uncharged, acidic, or basic and predict likely interactions with water or other side chains.
- Describe primary, secondary, tertiary, and quaternary protein structure and explain why a change in amino acid sequence can alter folding and function.

DNA RNA and the central dogma

- Compare DNA and RNA by sugar, bases, strand structure, stability, and cellular role.
- Use 5-prime and 3-prime directionality, antiparallel strands, and complementary base pairing. A pairs with T in DNA or U in RNA, and G pairs with C.
- Explain semiconservative DNA replication at the introductory level.
- Describe transcription from a DNA template into RNA and translation of mRNA by ribosomes and tRNAs at the introductory level.
- Use codons and anticodons, remember that a codon contains three nucleotides, and read a codon table.
- Connect a DNA sequence change to possible RNA, amino acid, protein, and trait consequences.

Cells chromosomes and inheritance

Review homologous chromosomes and the basic outcomes of cell division. Focus on how many cells result from mitosis and meiosis and how chromosome number changes.

- Describe a homologous chromosome pair, including how the chromosomes are related and how their alleles may differ.
- Compare the number of divisions and the number of cells produced by mitosis and meiosis.
- Explain how mitosis produces two daughter cells with the same chromosome number as the starting cell.
- Explain how meiosis produces four cells with half the chromosome number of the starting cell and connect this change to gamete production.

Genetics terms

- Review these terms and be able to use them correctly. Gene, allele, locus, genotype, phenotype, trait, mutation, wild type, homozygous, heterozygous, and carrier.
- Review how dominant and recessive describe relationships between alleles, including common misconceptions about those terms.
- Review how to use a Punnett square as a probability model and how to distinguish an individual's genotype from the probability of an offspring genotype.
- Review how Mendel's principles of segregation and independent assortment relate to meiosis.

Evolution and populations

- Review mutation and recombination, including how each affects genetic variation.
- Review how natural selection, genetic drift, and gene flow can change allele frequencies.
- Review the difference between changes that occur in populations across generations and changes that occur within individuals.

Ways to study

- Practice moving among words, diagrams, symbols, tables, and equations. For example, explain a Punnett square in a sentence and sketch meiosis from a verbal description.
- Draw from memory, label your drawing, then check it. Useful drawings include a nucleotide, an amino acid, a replicated chromosome, meiosis, DNA replication, transcription, and translation.
- Work problems without notes before checking the answer. If you miss one, identify whether the difficulty was vocabulary, the biological model, or the mathematics.
- When you answer a question, say why you chose the answer. If you cannot explain the biology behind it yet, that is a good topic to review.

A note about review in BIO 340

We will return to several of these topics in BIO 340 before using them in genetics problems. The review will be easier if the ideas already look familiar from BIO 181 and BIO 182. You do not need perfect recall.

Readiness check

Try these questions without notes or an internet search. If several feel unfamiliar, use the review plan on the next page and try again.

1. Convert $\frac{3}{8}$ to a decimal and a percentage.
2. A fair coin is flipped three times. What is the probability of three heads in a row?
3. Two heterozygous individuals, $Aa \times Aa$, are crossed. What is the probability of an aa offspring?
4. An organism has 18 chromosomes in a body cell. How many chromosomes should each daughter cell have after mitosis? How many should each resulting cell have after meiosis?
5. Describe a homologous chromosome pair. How can the two chromosomes in the pair differ?
6. A double-stranded DNA molecule contains 28% adenine. What percentages of thymine, guanine, and cytosine should it contain?
7. Name the covalent bond that joins nucleotides in a DNA strand and the interaction that pairs complementary bases across the two strands.
8. An amino acid side chain contains a carboxyl group that is negatively charged near physiological pH. Would you classify the side chain as nonpolar, polar uncharged, acidic, or basic?
9. Write the usual direction of information flow in the central dogma.
10. A DNA template strand reads 3-prime TAC GGA ACT 5-prime. Write the complementary mRNA sequence in the 5-prime to 3-prime direction.
11. Why is the statement A dominant allele is always the most common allele incorrect?
12. Does mutation, natural selection, genetic drift, or gene flow create a new allele? Explain briefly.

What to do with your results. Use the questions to choose what to refresh. The score is not a judgment about whether you belong in the course. If an answer is comfortable, move on. If it is uncertain, review that section in your BIO 181 or BIO 182 notes or an introductory biology text, then try the question again. You can access an open source biology textbook through OpenStax using [this link](#).

A five-session preparation plan

Review	Focus
Session 1	Fractions, decimals, percentages, ratios, probability, and graph reading. Finish with readiness questions 1 through 3.
Session 2	Homologous chromosomes, the number of cells produced by mitosis and meiosis, and changes in chromosome number. Finish with readiness questions 4 and 5.
Session 3	Chemical bonds, functional groups, nucleotides, amino acids, and protein structure. Finish with questions 7 and 8.
Session 4	DNA and RNA structure, replication, transcription, translation, codons, and mutations. Finish with questions 6, 9, and 10.
Session 5	Genetics vocabulary, Mendelian inheritance, and evolutionary processes. Finish with questions 11 and 12, then retake the entire readiness check without notes.

Readiness check answers

- 0.375 and 37.5%.
- $1/2 \times 1/2 \times 1/2 = 1/8$, or 12.5%.
- $1/4$, or 25%.
- Each daughter cell should have 18 chromosomes after mitosis. Each resulting cell should have 9 chromosomes after meiosis. Mitosis preserves chromosome number, while meiosis reduces it by half.
- A homologous pair usually includes one chromosome inherited from each parent. The chromosomes carry the same genes in the same locations, but they may carry different alleles.
- Thymine = 28%. The remaining 44% is divided equally between guanine and cytosine, so guanine = 22% and cytosine = 22%.
- Phosphodiester bonds join nucleotides within each strand. Hydrogen bonds pair complementary bases between strands.
- Acidic. The side chain is negatively charged and generally hydrophilic near physiological pH.
- DNA to RNA to protein. DNA can also be copied into DNA during replication.
- 5-prime AUG CCU UGA 3-prime.
- Dominance describes the phenotype of a heterozygote. It does not determine how frequent, beneficial, strong, or likely to be inherited an allele is.
- Mutation creates a new allele. Selection, drift, and gene flow can change the frequency of alleles already in a population. Gene flow may introduce an existing allele into a different population.